

# Exercises Day 1

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You already made exercises 1 and 2 in the slides. For exercises 3 to 7, you will do them in RStudio. Please make sure that you have R and RStudio installed on your laptop.

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## Exercise 1: selection within a vector

- Create a vector named `vec` that consists of the numbers from 11 to 30
- Select the 7th element of the vector
- Select all elements except the 15th. Hint: use a minus sign
- Select the 2nd and 5th element of the vector
- Select only the odd valued elements of `vec`.

```
# a.  
vec <- seq(11,30)  
# b.  
vec[7]
```

```
[1] 17
```

```
# c.  
vec[-15]
```

```
[1] 11 12 13 14 15 16 17 18 19 20 21 22 23 24 26 27 28 29 30
```

```
# d.  
vec[c(2,5)]
```

```
[1] 12 15
```

```
# e.  
index <- seq(1,length(vec),by=2)  
vec[index]
```

```
[1] 11 13 15 17 19 21 23 25 27 29
```

## Exercise 2: some calculations

- a. Use the elementary functions  $/$ ,  $-$ ,  $\wedge$  and the function `sum` to calculate the mean

$$\bar{x} = \sum_{i=1}^n x_i / n$$

and the standard deviation

$$\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 / (n - 1)}$$

of the fare paid by the titanic passengers.

- b. Verify your answer by using the built-in R functions `mean` and `sd`.

We create the data set that we used in the web based interface.

```
titanic <- data.frame(
  pclass=rep(c("1st","2nd","3rd"),c(4,2,4)),
  survived=c(1,0,0,0,1,1,0,0,0,0),
  name=c("Allison, Master. Hudson Trevor","Dulles, Mr. William Crothers",
    "McCarthy, Mr. Timothy J","Walker, Mr. William Anderson",
    "Duran y More, Miss. Asuncion", "Mellinger, Miss. Madeleine Viol",
    "Abbott, Master. Eugene Joseph", "Calic, Mr. Petar","Flynn, Mr. James",
    "Johnston, Miss. Catherine Helen"),
  age= c(0.9167,39,54,47,27,13,13,17,NA,NA),
  fare=c(151.55,29.7,51.8625,34.0208,13.8583,19.5,20.25,8.6625,7.75,23.4))
```

```
# a.
fare <- titanic$fare
MeanFare <- sum(fare)/length(fare)
MeanFare
```

```
[1] 36.05541
```

```
# standard deviation (in three baby steps)
numerator <- sum((fare-MeanFare)^2)
denominator <- (length(fare)-1)
StdFare <- sqrt(numerator/denominator)
StdFare
```

```
[1] 42.63766
```

```
# b.
mean(fare)
```

```
[1] 36.05541
```

```
sd(fare)
```

```
[1] 42.63766
```

## RStudio

### Some useful Keyboard Shortcuts in RStudio for Windows

Over time it is useful to learn a couple of RStudio shortcuts that you frequently use. You can find an overview of them under the **Help** → **Keyboard Shortcuts help** menu. More shortcuts and tips can be found on the [Apsilon](#) website.

#### Note

On the mac, Ctrl → Cmd, Alt → Option, and Enter → Return.

- **Ctrl+Enter**: Run current line
- **Ctrl+Alt+I**: Insert new chunk
- **Alt+-**: Insert assignment <-
- **F1**: Show function help page when cursor is on function name
- **Ctrl+Tab**: Switch between tabs in Source Editor
- **Ctrl+1** / **Ctrl+2**: Move focus to Source/Console
- **Tab** / **Ctrl+Space**: Code suggestion/completion
- **In Console**:
  - Arrows **Up/Down** to navigate earlier commands
  - **Ctrl+Up** pops up a listing of latest commands

Create a folder on your computer for the exercises in this R course. Open RStudio and create a new New Project in that folder. Open a new R Script file via the menu:

**File** → **New File** → **R Script...**

Write the code in that R Script file.

### Exercise 3: importing data from Excel and first check

We will work with a dataset that gives the survival status of passengers on the Titanic. We provide the dataset in **Excel** format ([Titanic3.xlsx](#)). See [this link](#) for a description of the variables. Our data set has three additional columns: **dob** (date of birth), **family** (whether the person travelled with family) and **agecat** (age categorized in age ranges).

Download the dataset to the raw data subfolder of your project folder. Import the dataset by clicking on **Import Dataset** from the **Environment** pane. Browse for the Excel file.

The first 50 rows of the data set are shown. Look at the code under **Code Preview**. Assign the R object name `titanic` to the dataset under **Import Options** and observe that the code under **Code Preview** has changed.

Look at the mode of variable `age`. A problem with the mode for `age` is that missing values are coded as a character value "NA" in our Excel file; it is not recognized as an R type NA value. In the future, you may encounter other representations of missing values, such as `missing`, `404`, `-99` or `unknown`, which should also be treated as NA. Specify that NA represents missing values under the import options. Observe how the code under **\*Code Preview changed again. Copy the code from Code Preview\***. Click on the "Import" button. Paste the code you have copied earlier to your R Script File.

- a. Inspect the data set in spreadsheet-like format that shows up after importing (if it does not show up, you can click on the `titanic` object in the **Environment** pane). Do the variables make sense? Are there any weird values? Sort the data in ascending order of `age` by clicking on **age** in the data set.

**Answer:** The variable `dob` has a strange format. The variable `survived` has values 0 and 1, which are not very informative.

- b. Use the function `dim` to find the number of rows (passengers) and columns (variables) in the data set.

```
dim(titanic)
```

```
[1] 1309  17
```

**Answer:** 1309 rows and 17 columns

- c. Use the `str` function to obtain a summary of the basic characteristics per variable. Have a look at the `survived` column. Does it have the mode that you expect? (You can also use the `mode` function to check the modes.)

```
str(titanic)
```

```
tibble [1,309 x 17] (S3: tbl_df/tbl/data.frame)
 $ pclass   : chr [1:1309] "1st" "1st" "1st" "1st" ...
 $ survived : num [1:1309] 1 1 0 0 0 1 1 0 1 0 ...
 $ name     : chr [1:1309] "Allen, Miss. Elisabeth Walton" "Allison, Master. Hudson Tr
 $ sex      : chr [1:1309] "female" "male" "female" "male" ...
 $ age      : num [1:1309] 29 0.917 2 30 25 ...
```

```

$ sibsp      : num [1:1309] 0 1 1 1 1 0 1 0 2 0 ...
$ parch      : num [1:1309] 0 2 2 2 2 0 0 0 0 0 ...
$ ticket     : chr [1:1309] "24160" "113781" "113781" "113781" ...
$ fare       : num [1:1309] 211 152 152 152 152 ...
$ cabin      : chr [1:1309] "B5" "C22 C26" "C22 C26" "C22 C26" ...
$ embarked   : chr [1:1309] "Southampton" "Southampton" "Southampton" "Southampton" ...
$ boat       : chr [1:1309] "2" "11" NA NA ...
$ body       : num [1:1309] NA NA NA 135 NA NA NA NA NA 22 ...
$ home_dest  : chr [1:1309] "St Louis, MO" "Montreal, PQ / Chesterville, ON" "Montreal,
$ dob        : num [1:1309] -28019 -17762 -18158 -28384 -26558 ...
$ family     : chr [1:1309] "no" "yes" "yes" "yes" ...
$ agecat     : chr [1:1309] "(18,30]" "(0,18]" "(0,18]" "(18,30]" ...

```

```
mode(titanic$survived)
```

```
[1] "numeric"
```

survived is of mode numeric, while one would expect it to be a categorical variable of mode character.

## Exercise 4: some columns summarized

a. Issue the commands

```
head(titanic, 10)
```

```

# A tibble: 10 x 17
  pclass survived name      sex      age sibsp parch ticket  fare cabin embarked
  <chr>      <dbl> <chr>    <chr>    <dbl> <dbl> <dbl> <chr>  <dbl> <chr> <chr>
1 1st             1 Allen, ~ fema~ 29             0      0 24160  211.  B5    Southam~
2 1st             1 Allison~ male   0.917          1      2 113781 152.  C22 ~ Southam~
3 1st             0 Allison~ fema~ 2             1      2 113781 152.  C22 ~ Southam~
4 1st             0 Allison~ male   30             1      2 113781 152.  C22 ~ Southam~
5 1st             0 Allison~ fema~ 25             1      2 113781 152.  C22 ~ Southam~
6 1st             1 Anderso~ male   48             0      0 19952  26.5 E12    Southam~
7 1st             1 Andrews~ fema~ 63             1      0 13502  78.0 D7     Southam~
8 1st             0 Andrews~ male   39             0      0 112050  0    A36    Southam~
9 1st             1 Appleto~ fema~ 53             2      0 11769  51.5 C101   Southam~

```

```
10 1st          0 Artagav~ male  71          0      0 PC 17~ 49.5 <NA> Cherbou~
# i 6 more variables: boat <chr>, body <dbl>, home_dest <chr>, dob <dbl>,
#   family <chr>, agecat <chr>
```

```
tail(titanic, 10)
```

```
# A tibble: 10 x 17
  pclass survived name      sex    age sibsp parch ticket  fare cabin embarked
  <chr>      <dbl> <chr>      <chr> <dbl> <dbl> <dbl> <chr>  <dbl> <chr> <chr>
1 3rd          0 Yasbeck,~ male   27      1      0 2659  14.5 <NA> Cherbou~
2 3rd          1 Yasbeck,~ fema~  15      1      0 2659  14.5 <NA> Cherbou~
3 3rd          0 Yousseff,~ male  45.5      0      0 2628   7.22 <NA> Cherbou~
4 3rd          0 Yousif, ~ male   NA      0      0 2647   7.22 <NA> Cherbou~
5 3rd          0 Yousseff~ male   NA      0      0 2627  14.5 <NA> Cherbou~
6 3rd          0 Zabour, ~ fema~  14.5      1      0 2665  14.5 <NA> Cherbou~
7 3rd          0 Zabour, ~ fema~   NA      1      0 2665  14.5 <NA> Cherbou~
8 3rd          0 Zakarian~ male  26.5      0      0 2656   7.22 <NA> Cherbou~
9 3rd          0 Zakarian~ male   27      0      0 2670   7.22 <NA> Cherbou~
10 3rd          0 Zimmerma~ male   29      0      0 315082  7.88 <NA> Southam~
# i 6 more variables: boat <chr>, body <dbl>, home_dest <chr>, dob <dbl>,
#   family <chr>, agecat <chr>
```

What do you observe?

**Answer:** First and last 10 rows are shown. Not all columns are shown.

- b. The `readxl` package that is used to import the data stores the data as a **tibble**, which is a special type of format on top of the `data.frame` format to store data in R. Override the **tibble** format using the function `as.data.frame`.

```
titanic <- as.data.frame(titanic)
```

Run the `head` and `tail` functions again. What has changed?

```
head(titanic, 10)
```

```
  pclass survived      name      sex    age sibsp parch
1     1st         1 Allen, Miss. Elisabeth Walton female 29.0000      0      0
2     1st         1 Allison, Master. Hudson Trevor   male  0.9167      1      2
```

|    |                |          |                                 |             |         |         |   |
|----|----------------|----------|---------------------------------|-------------|---------|---------|---|
| 3  | 1st            | 0        | Allison, Miss. Helen Loraine    | female      | 2.0000  | 1       | 2 |
| 4  | 1st            | 0        | Allison, Mr. Hudson Joshua Crei | male        | 30.0000 | 1       | 2 |
| 5  | 1st            | 0        | Allison, Mrs. Hudson J C (Bessi | female      | 25.0000 | 1       | 2 |
| 6  | 1st            | 1        | Anderson, Mr. Harry             | male        | 48.0000 | 0       | 0 |
| 7  | 1st            | 1        | Andrews, Miss. Kornelia Theodos | female      | 63.0000 | 1       | 0 |
| 8  | 1st            | 0        | Andrews, Mr. Thomas Jr          | male        | 39.0000 | 0       | 0 |
| 9  | 1st            | 1        | Appleton, Mrs. Edward Dale (Cha | female      | 53.0000 | 2       | 0 |
| 10 | 1st            | 0        | Artagaveytia, Mr. Ramon         | male        | 71.0000 | 0       | 0 |
|    | ticket         | fare     | cabin                           | embarked    | boat    | body    |   |
| 1  | 24160          | 211.3375 | B5                              | Southampton | 2       | NA      |   |
| 2  | 113781         | 151.5500 | C22 C26                         | Southampton | 11      | NA      |   |
| 3  | 113781         | 151.5500 | C22 C26                         | Southampton | <NA>    | NA      |   |
| 4  | 113781         | 151.5500 | C22 C26                         | Southampton | <NA>    | 135     |   |
| 5  | 113781         | 151.5500 | C22 C26                         | Southampton | <NA>    | NA      |   |
| 6  | 19952          | 26.5500  | E12                             | Southampton | 3       | NA      |   |
| 7  | 13502          | 77.9583  | D7                              | Southampton | 10      | NA      |   |
| 8  | 112050         | 0.0000   | A36                             | Southampton | <NA>    | NA      |   |
| 9  | 11769          | 51.4792  | C101                            | Southampton | D       | NA      |   |
| 10 | PC 17609       | 49.5042  | <NA>                            | Cherbourg   | <NA>    | 22      |   |
|    |                |          | home_dest                       | dob         | family  | agecat  |   |
| 1  |                |          | St Louis, MO                    | -28019.25   | no      | (18,30] |   |
| 2  | Montreal, PQ / |          | Chesterville, ON                | -17761.82   | yes     | (0,18]  |   |
| 3  | Montreal, PQ / |          | Chesterville, ON                | -18157.50   | yes     | (0,18]  |   |
| 4  | Montreal, PQ / |          | Chesterville, ON                | -28384.50   | yes     | (18,30] |   |
| 5  | Montreal, PQ / |          | Chesterville, ON                | -26558.25   | yes     | (18,30] |   |
| 6  |                |          | New York, NY                    | -34959.00   | no      | (40,80] |   |
| 7  |                |          | Hudson, NY                      | -40437.75   | yes     | (40,80] |   |
| 8  |                |          | Belfast, NI                     | -31671.75   | no      | (30,40] |   |
| 9  |                |          | Bayside, Queens, NY             | -36785.25   | yes     | (40,80] |   |
| 10 |                |          | Montevideo, Uruguay             | -43359.75   | no      | (40,80] |   |

```
tail(titanic, 10)
```

|      | pclass | survived | name                            | sex    | age  | sibsp | parch |
|------|--------|----------|---------------------------------|--------|------|-------|-------|
| 1300 | 3rd    | 0        | Yasbeck, Mr. Antoni             | male   | 27.0 | 1     | 0     |
| 1301 | 3rd    | 1        | Yasbeck, Mrs. Antoni (Selini Al | female | 15.0 | 1     | 0     |
| 1302 | 3rd    | 0        | Youseff, Mr. Gerious            | male   | 45.5 | 0     | 0     |
| 1303 | 3rd    | 0        | Yousif, Mr. Wazli               | male   | NA   | 0     | 0     |
| 1304 | 3rd    | 0        | Yousseff, Mr. Gerious           | male   | NA   | 0     | 0     |
| 1305 | 3rd    | 0        | Zabour, Miss. Hileni            | female | 14.5 | 1     | 0     |

|      |     |   |  |                           |        |      |   |   |
|------|-----|---|--|---------------------------|--------|------|---|---|
| 1306 | 3rd | 0 |  | Zabour, Miss. Thamine     | female | NA   | 1 | 0 |
| 1307 | 3rd | 0 |  | Zakarian, Mr. Mapriededer | male   | 26.5 | 0 | 0 |
| 1308 | 3rd | 0 |  | Zakarian, Mr. Ortin       | male   | 27.0 | 0 | 0 |
| 1309 | 3rd | 0 |  | Zimmerman, Mr. Leo        | male   | 29.0 | 0 | 0 |

  

|      | ticket | fare    | cabin | embarked    | boat | body | home_dest | dob       | family |
|------|--------|---------|-------|-------------|------|------|-----------|-----------|--------|
| 1300 | 2659   | 14.4542 | <NA>  | Cherbourg   | C    | NA   | <NA>      | -27288.75 | yes    |
| 1301 | 2659   | 14.4542 | <NA>  | Cherbourg   | <NA> | NA   | <NA>      | -22905.75 | yes    |
| 1302 | 2628   | 7.2250  | <NA>  | Cherbourg   | <NA> | 312  | <NA>      | -34045.88 | no     |
| 1303 | 2647   | 7.2250  | <NA>  | Cherbourg   | <NA> | NA   | <NA>      | NA        | no     |
| 1304 | 2627   | 14.4583 | <NA>  | Cherbourg   | <NA> | NA   | <NA>      | NA        | no     |
| 1305 | 2665   | 14.4542 | <NA>  | Cherbourg   | <NA> | 328  | <NA>      | -22723.12 | yes    |
| 1306 | 2665   | 14.4542 | <NA>  | Cherbourg   | <NA> | NA   | <NA>      | NA        | yes    |
| 1307 | 2656   | 7.2250  | <NA>  | Cherbourg   | <NA> | 304  | <NA>      | -27106.12 | no     |
| 1308 | 2670   | 7.2250  | <NA>  | Cherbourg   | <NA> | NA   | <NA>      | -27288.75 | no     |
| 1309 | 315082 | 7.8750  | <NA>  | Southampton | <NA> | NA   | <NA>      | -28019.25 | no     |

  

|      | agecat  |
|------|---------|
| 1300 | (18,30] |
| 1301 | (0,18]  |
| 1302 | (40,80] |
| 1303 | <NA>    |
| 1304 | <NA>    |
| 1305 | (0,18]  |
| 1306 | <NA>    |
| 1307 | (18,30] |
| 1308 | (18,30] |
| 1309 | (18,30] |

**Answer:** All columns are shown.

- c. Summarize the whole dataset using the `summary` function. What do you observe for each of the variables?

```
summary(titanic)
```

| pclass    |       | survived |        | name      |       | sex       |       |
|-----------|-------|----------|--------|-----------|-------|-----------|-------|
| Length    | :1309 | Min.     | :0.000 | Length    | :1309 | Length    | :1309 |
| N.unique  | : 3   | 1st Qu.  | :0.000 | N.unique  | :1307 | N.unique  | : 2   |
| N.blank   | : 0   | Median   | :0.000 | N.blank   | : 0   | N.blank   | : 0   |
| Min.nchar | : 3   | Mean     | :0.382 | Min.nchar | : 12  | Min.nchar | : 4   |
| Max.nchar | : 3   | 3rd Qu.  | :1.000 | Max.nchar | : 33  | Max.nchar | : 6   |

Max. :1.000

| age             | sibsp          | parch         | ticket         |
|-----------------|----------------|---------------|----------------|
| Min. : 0.1667   | Min. :0.0000   | Min. :0.000   | Length :1309   |
| 1st Qu.:21.0000 | 1st Qu.:0.0000 | 1st Qu.:0.000 | N.unique : 929 |
| Median :28.0000 | Median :0.0000 | Median :0.000 | N.blank : 0    |
| Mean :29.8811   | Mean :0.4989   | Mean :0.385   | Min.nchar: 3   |
| 3rd Qu.:39.0000 | 3rd Qu.:1.0000 | 3rd Qu.:0.000 | Max.nchar: 18  |
| Max. :80.0000   | Max. :8.0000   | Max. :9.000   |                |
| NAs :263        |                |               |                |

  

| fare           | cabin          | embarked      | boat          |
|----------------|----------------|---------------|---------------|
| Min. : 0.000   | Length :1309   | Length :1309  | Length :1309  |
| 1st Qu.: 7.896 | N.unique : 186 | N.unique : 3  | N.unique : 27 |
| Median :14.454 | N.blank : 0    | N.blank : 0   | N.blank : 0   |
| Mean : 33.295  | Min.nchar: 1   | Min.nchar: 9  | Min.nchar: 1  |
| 3rd Qu.:31.275 | Max.nchar: 15  | Max.nchar: 11 | Max.nchar: 7  |
| Max. :512.329  | NAs :1014      | NAs : 2       | NAs : 823     |
| NAs :1         |                |               |               |

  

| body          | home_dest      | dob             | family       |
|---------------|----------------|-----------------|--------------|
| Min. : 1.0    | Length :1309   | Min. : -46647   | Length :1309 |
| 1st Qu.:72.0  | N.unique : 368 | 1st Qu.: -31672 | N.unique : 2 |
| Median :155.0 | N.blank : 0    | Median : -27654 | N.blank : 0  |
| Mean :160.8   | Min.nchar: 5   | Mean : -28341   | Min.nchar: 2 |
| 3rd Qu.:256.0 | Max.nchar: 31  | 3rd Qu.: -25097 | Max.nchar: 3 |
| Max. :328.0   | NAs : 564      | Max. : -17488   |              |
| NAs :1188     |                | NAs :263        |              |

  

| agecat       |
|--------------|
| Length :1309 |
| N.unique : 4 |
| N.blank : 0  |
| Min.nchar: 6 |
| Max.nchar: 7 |
| NAs : 263    |

Many are of class and mode character, and the `summary` function does not give any information. The `dob` variable has negative values.

- d. Compute the 5%, 25%, 50%, 75% and 95% quantiles, the inter-quartile range and the standard deviation for age and fare.

```
## some quantiles
quantile(titanic$age, probs=c(0.05,0.25,0.5,0.75,0.95), na.rm=TRUE)
```

```
5% 25% 50% 75% 95%
5  21  28  39  57
```

```
quantile(titanic$fare, probs=c(0.05,0.25,0.5,0.75,0.95), na.rm=TRUE)
```

```
      5%      25%      50%      75%      95%
7.2250  7.8958 14.4542 31.2750 133.6500
```

```
## the IQR (note that this is a single number)
IQR(titanic$age, na.rm=TRUE)
```

```
[1] 18
```

```
IQR(titanic$fare, na.rm=TRUE)
```

```
[1] 23.3792
```

```
## the the standard deviation
sd(titanic$age, na.rm = TRUE)
```

```
[1] 14.4135
```

```
sd(titanic$fare, na.rm = TRUE)
```

```
[1] 51.75867
```

- e. For categorical variables, `summary` is not very informative. Use the `table` function to summarize the variables `pclass`, and `sex`. Also give a two-by-two table for sex and survival status using the same `table` function. Again, have a look at the help file if needed.

```
## basic tables
table(titanic$pclass)
```

```
1st 2nd 3rd
323 277 709
```

```
table(titanic$sex)
```

```
female  male
    466    843
```

```
table(titanic$sex, titanic$survived)
```

```
      0    1
female 127 339
male   682 161
```

```
## argument useNA has no added value because there are no missings in
## sex and survived
table(titanic$sex, titanic$survived, useNA = "always")
```

```
      0    1 <NA>
female 127 339    0
male   682 161    0
<NA>    0    0    0
```

- f. The function `addmargins` adds the row sums and the column sums to the table of survival status by sex. The function `proportions` tabulates proportions instead of absolute numbers.

```
addmargins(table(titanic$sex, titanic$survived))
```

```
      0    1 Sum
female 127 339 466
male   682 161 843
Sum    809 500 1309
```

```
proportions(table(titanic$sex, titanic$survived))
```

|        | 0          | 1          |
|--------|------------|------------|
| female | 0.09702063 | 0.25897632 |
| male   | 0.52100840 | 0.12299465 |

Use the `proportions` function to compute the fraction (or percentage) of survivors by sex.

```
## basic tables
proportions(table(titanic$sex, titanic$survived), margin = 1)
```

|        | 0         | 1         |
|--------|-----------|-----------|
| female | 0.2725322 | 0.7274678 |
| male   | 0.8090154 | 0.1909846 |

## Exercise 5: work with subsets

- Take a look at the name, and the home town/destination of all passengers who were older than 70 years. Use the appropriate selection functions.

```
subset(titanic, age > 70)[, c("name", "home_dest")]
```

|      | name                                            | home_dest           |
|------|-------------------------------------------------|---------------------|
| 10   | Artagaveytia, Mr. Ramon                         | Montevideo, Uruguay |
| 15   | Barkworth, Mr. Algernon Henry W                 | Hessle, Yorks       |
| 62   | Cavendish, Mrs. Tyrell William Little Onn Hall, | Staffs              |
| 136  | Goldschmidt, Mr. George B                       | New York, NY        |
| 728  | Connors, Mr. Patrick                            | <NA>                |
| 1236 | Svensson, Mr. Johan                             | <NA>                |

It is seen that there is one person from Uruguay in this group. Select the single record from that person. Did this person travel with relatives?

```
subset(titanic, name == "Artagaveytia, Mr. Ramon")
```

```

      pclass survived      name sex age sibsp parch ticket
10      1st          0 Artagaveytia, Mr. Ramon male  71      0      0 PC 17609
      fare cabin embarked boat body      home_dest      dob family
10 49.5042 <NA> Cherbourg <NA>  22 Montevideo, Uruguay -43359.75      no
      agecat
10 (40,80]

```

**Answer:** the person did not travel with relatives

- b. Make a table of survivor status by sex, but only for the first class passengers. What do you observe? Compare this with the survival status for the 3rd class passengers.

```

first <- xtabs(~sex+survived, data=titanic, subset=(pclass=="1st"))
first

```

```

      survived
sex      0    1
  female  5 139
   male 118  61

```

```

third <- xtabs(~sex+survived, data=titanic, subset=(pclass=="3rd"))
third

```

```

      survived
sex      0    1
  female 110 106
   male  418  75

```

**Answer:** females were much more likely to survive than males, especially among those who travelled first class

## Exercise 6: make factor variables

- Make passenger class and sex into factor variables.
- Add a variable `status` to the Titanic data set, which gives the survival status of the passengers as a factor variable with labels “no” and “yes” describing whether individuals survived. Make the value “no” the first level.

```
titanic$status <- factor(titanic$survived, labels=c("no","yes"))
```

- c. Run the `summary` function again. What change do you observe compared to Exercise 4?

```
summary(titanic)
```

| pclass     |       | survived |        | name       |       | sex        |       |
|------------|-------|----------|--------|------------|-------|------------|-------|
| Length     | :1309 | Min.     | :0.000 | Length     | :1309 | Length     | :1309 |
| N.unique   | : 3   | 1st Qu.: | :0.000 | N.unique   | :1307 | N.unique   | : 2   |
| N.blank    | : 0   | Median   | :0.000 | N.blank    | : 0   | N.blank    | : 0   |
| Min.nchar: | 3     | Mean     | :0.382 | Min.nchar: | 12    | Min.nchar: | 4     |
| Max.nchar: | 3     | 3rd Qu.: | :1.000 | Max.nchar: | 33    | Max.nchar: | 6     |
|            |       | Max.     | :1.000 |            |       |            |       |

  

| age      |          | sibsp    |         | parch    |        | ticket     |       |
|----------|----------|----------|---------|----------|--------|------------|-------|
| Min.     | : 0.1667 | Min.     | :0.0000 | Min.     | :0.000 | Length     | :1309 |
| 1st Qu.: | :21.0000 | 1st Qu.: | :0.0000 | 1st Qu.: | :0.000 | N.unique   | : 929 |
| Median   | :28.0000 | Median   | :0.0000 | Median   | :0.000 | N.blank    | : 0   |
| Mean     | :29.8811 | Mean     | :0.4989 | Mean     | :0.385 | Min.nchar: | 3     |
| 3rd Qu.: | :39.0000 | 3rd Qu.: | :1.0000 | 3rd Qu.: | :0.000 | Max.nchar: | 18    |
| Max.     | :80.0000 | Max.     | :8.0000 | Max.     | :9.000 |            |       |
| NAs      | :263     |          |         |          |        |            |       |

  

| fare     |          | cabin      |       | embarked   |       | boat       |       |
|----------|----------|------------|-------|------------|-------|------------|-------|
| Min.     | : 0.000  | Length     | :1309 | Length     | :1309 | Length     | :1309 |
| 1st Qu.: | 7.896    | N.unique   | : 186 | N.unique   | : 3   | N.unique   | : 27  |
| Median   | :14.454  | N.blank    | : 0   | N.blank    | : 0   | N.blank    | : 0   |
| Mean     | : 33.295 | Min.nchar: | 1     | Min.nchar: | 9     | Min.nchar: | 1     |
| 3rd Qu.: | 31.275   | Max.nchar: | 15    | Max.nchar: | 11    | Max.nchar: | 7     |
| Max.     | :512.329 | NAs        | :1014 | NAs        | : 2   | NAs        | : 823 |
| NAs      | :1       |            |       |            |       |            |       |

  

| body     |        | home_dest  |       | dob      |          | family     |       |
|----------|--------|------------|-------|----------|----------|------------|-------|
| Min.     | : 1.0  | Length     | :1309 | Min.     | : -46647 | Length     | :1309 |
| 1st Qu.: | 72.0   | N.unique   | : 368 | 1st Qu.: | : -31672 | N.unique   | : 2   |
| Median   | :155.0 | N.blank    | : 0   | Median   | : -27654 | N.blank    | : 0   |
| Mean     | :160.8 | Min.nchar: | 5     | Mean     | : -28341 | Min.nchar: | 2     |
| 3rd Qu.: | 256.0  | Max.nchar: | 31    | 3rd Qu.: | : -25097 | Max.nchar: | 3     |
| Max.     | :328.0 | NAs        | : 564 | Max.     | : -17488 |            |       |
| NAs      | :1188  |            |       | NAs      | :263     |            |       |

  

| agecat |  | status |  |
|--------|--|--------|--|
|--------|--|--------|--|

```

Length    :1309    no :809
N.unique  :   4    yes:500
N.blank   :   0
Min.nchar :   6
Max.nchar :   7
NAs       : 263

```

**Answer:** Passenger class and sex are summarized properly, and status is summarized as a categorical variable.

## Exercise 7: working with dates

- The variable `dob` gives the date of birth, using days since January 1st, 1960 (this is the time origin used by Stata). Transform this to a true date value using the function `as.Date` (specify the origin argument).

```
titanic$dob <- as.Date(titanic$dob, origin = "1960/1/1")
```

- The default display is “year/month/day”. Make a table of the day of the month that the passengers were born, using the `format` function. What do you observe?

```
table(format(titanic$dob, "%d"))
```

```

13  14  15  16
15 940  86   5

```

**Answer:** All have a day in the middle of the month

- What was the earliest date of birth among all the passengers on the boat. Does this correspond to the age of the oldest person on the date that the ship sank (April 15th, 1912)? Use the functions `min` and `max`.

```
min(titanic$dob, na.rm = TRUE)
```

```
[1] "1832-04-14"
```

```
max(titanic$age, na.rm = TRUE)
```

```
[1] 80
```

```
min(titanic$dob, na.rm = TRUE) + max(titanic$age, na.rm = TRUE) * 365.25
```

```
[1] "1912-04-15"
```

Yes, it does correspond